

Supplementary Material for Theorem 36
*On the Minimum Distance of High-rate Moore-Type
 Quasi-Cyclic LDPC Codes*

Detailed Derivations for Weight-8 Cases II–VII

1 Scope

This document supplies the detailed calculations for Cases II–VII in Theorem 36 of the main manuscript, which characterizes Hamming-weight-8 codewords of Moore($m, a, b, 3$) when $\text{ord}_m(b) = 3$. Case I is proved in full in the main manuscript. The notation, normalization, case labels, and conditions (a)–(d) are unchanged.

The correspondence between the case labels used below and the matching subcases is

Case	Matching subcase	RREF	Free variables
II	1-2-1	A_{121}	u_4, u_7, u_8
III	1-2-2	A_{122}	u_6, u_7, u_8
IV	2-1-2	A_{212}	u_4, u_6, u_7, u_8
V	2-1-7	A_{217}	u_4, u_6, u_7, u_8
VI	2-3-5	A_{235}	u_4, u_6, u_7, u_8
VII	2-4-3	A_{243}	u_5, u_6, u_7, u_8

2 Algebraic framework and conventions

All calculations are carried out in the finite field $\mathbb{F}_m$. We use the normalization $c_1 = (0, 0)$ and write

$$u_i = a^{t_i}, \quad u_1 = 1.$$

Since $\text{ord}_m(b) = 3$, we have $b \neq 1$ and

$$1 + b + b^2 = 0.$$

All coefficients below are simplified using this identity.

For a matching edge $\{c_i, c_q\}$ at level $s \in \{0, 1, 2\}$, the basic relation is

$$x_i - x_q = (b^s - 1)(u_q - u_i).$$

For an oriented cycle

$$\underline{i_1}, i_2, \dots, i_\ell (s_1 s_2 \cdots s_\ell),$$

the underlined indices denote the successive vertices and s_r is the level of the edge joining c_{i_r} to $c_{i_{r+1}}$, with $i_{\ell+1} = i_1$. Summing the basic edge relations around the cycle eliminates the x -coordinates and gives a homogeneous linear equation in $u_1, \dots, u_8$.

For each configuration, choose a spanning tree and form the fundamental-cycle equations. These equations are sufficient as well as necessary: the tree-edge relations determine all x_i from

one root value, and each non-tree edge is compatible with them exactly when its fundamental-cycle equation vanishes. We then row-reduce in the fixed column order $(u_1, \dots, u_8)$, omitting zero rows. After $u_1 = 1$, the pivot relation for u_1 gives condition (b), and the remaining dependent variables give condition (d) through membership in $\langle a \rangle$.

For distinctness, the level-0 blocks must have different common x -coordinates, while columns inside the same level-0 block must have different exponents. Substituting the RREF relations and the main congruence into all required differences gives conditions (a) and (c). Factors differing by a nonzero scalar in $\mathbb{F}_m$ are identified.

Labeling convention for Case VI. For subcase 2-3-5, we use the representative in Proposition 29. Relative to the intermediate enumeration label, it is obtained by $c_7 \leftrightarrow c_8$; this fixes the column order used for A_{235} .

3 Case II: Subcase 1-2-1

3.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3, c_6, c_8\}, \{c_2, c_5\}, \{c_4, c_7\}\}, \\ M_2 &= \{\{c_1, c_4\}, \{c_2, c_3\}, \{c_5, c_8\}, \{c_6, c_7\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_1, c_6\}_1, \{c_1, c_8\}_1, \{c_2, c_5\}_1, \{c_4, c_7\}_1, \{c_1, c_4\}_2, \{c_2, c_3\}_2.$$

This tree has 7 edges. Since the matching graph has 16 edges, it determines 9 fundamental cycles. The associated cycle equations are listed below. Three fundamental cycles, $\underline{3, 1, 6}(111)$, $\underline{3, 1, 8}(111)$, and $\underline{6, 1, 8}(111)$, lie entirely in the level-1 4-matching and give the identity $0 = 0$. The six nontrivial fundamental-cycle equations are:

(1) $\underline{5, 2, 3, 1, 8}(12112)$:

$$(1 + 2b)u_2 + (-1 - 2b)u_3 + (-1 - 2b)u_5 + (1 + 2b)u_8 = 0.$$

(2) $\underline{6, 1, 4, 7}(1212)$:

$$(1 + 2b)u_1 + (-1 - 2b)u_4 + (-1 - 2b)u_6 + (1 + 2b)u_7 = 0.$$

(3) $\underline{1, 3, 2}(120)$:

$$(1 - b)u_1 + (-2 - b)u_2 + (1 + 2b)u_3 = 0.$$

(4) $\underline{3, 1, 4}(120)$:

$$(1 + 2b)u_1 + (1 - b)u_3 + (-2 - b)u_4 = 0.$$

(5) $\underline{5, 2, 3, 1, 6}(12110)$:

$$(1 + 2b)u_2 + (-1 - 2b)u_3 + (1 - b)u_5 + (-1 + b)u_6 = 0.$$

(6) $\underline{7, 4, 1, 8}(1210)$:

$$(-1 - 2b)u_1 + (1 + 2b)u_4 + (1 - b)u_7 + (-1 + b)u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{121} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 1+b & -1-b \\ 0 & 1 & 0 & -1 & 0 & 0 & 2+b & -2-b \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1+b & -2-b \\ 0 & 0 & 0 & 0 & 0 & 1 & b & -1-b \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

3.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_4 + (-1-b)u_7 + (1+b)u_8, \\ u_2 &= u_4 + (-2-b)u_7 + (2+b)u_8, \\ u_3 &= u_4 - u_7 + u_8, \\ u_5 &= (-1-b)u_7 + (2+b)u_8, \\ u_6 &= -bu_7 + (1+b)u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + (-1-b)u_7 + (1+b)u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d) .

3.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_4 :

$$u_4 = 1 + (1+b)u_7 + (-1-b)u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (1+2b)u_7 + (-1-2b)u_8, \\ x_5 &= x_6 = -1+b + (-1-2b)u_7 + (2+b)u_8, \\ x_7 &= x_8 = -1+b + (1-b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_7 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + bu_7 + (-1-b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + 2bu_7 + (-1-2b)u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_8 \neq 0 &\iff 1 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 + bu_7 + (-1 - b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + 2bu_7 + (-1 - 2b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

4 Case III: Subcase 1-2-2

4.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4, c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_5\}, \{c_4, c_7\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_4\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_5, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_5\}_1, \{c_4, c_7\}_1, \{c_6, c_8\}_1, \{c_1, c_4\}_2, \{c_2, c_6\}_2, \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 16 edges, it determines 9 fundamental cycles. The associated cycle equations are listed below.

(1) 3, 1, 4, 7(1212):

$$(1 + 2b)u_1 + (-1 - 2b)u_3 + (-1 - 2b)u_4 + (1 + 2b)u_7 = 0.$$

(2) 5, 2, 6, 8(1212):

$$(1 + 2b)u_2 + (-1 - 2b)u_5 + (-1 - 2b)u_6 + (1 + 2b)u_8 = 0.$$

(3) 3, 1, 4(120):

$$(1 + 2b)u_1 + (1 - b)u_3 + (-2 - b)u_4 = 0.$$

(4) 3, 1, 2, 5(1010):

$$(-1 + b)u_1 + (1 - b)u_2 + (1 - b)u_3 + (-1 + b)u_5 = 0.$$

(5) 3, 1, 2, 6(1020):

$$(-1 + b)u_1 + (2 + b)u_2 + (1 - b)u_3 + (-2 - b)u_6 = 0.$$

(6) 4, 1, 2, 5(2010):

$$(-2 - b)u_1 + (1 - b)u_2 + (2 + b)u_4 + (-1 + b)u_5 = 0.$$

(7) 4, 1, 2, 6(2020):

$$(-2 - b)u_1 + (2 + b)u_2 + (2 + b)u_4 + (-2 - b)u_6 = 0.$$

(8) 5, 2, 6(120):

$$(1 + 2b) u_2 + (1 - b) u_5 + (-2 - b) u_6 = 0.$$

(9) 7, 4, 1, 2, 6, 8(120210):

$$\begin{aligned} &(-2 - b) u_1 + (2 + b) u_2 + (1 + 2b) u_4 + (-1 - 2b) u_6 \\ &+ (1 - b) u_7 + (-1 + b) u_8 = 0. \end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{122} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1+b & -1 & 1-b \\ 0 & 1 & 0 & 0 & 0 & -1+b & 0 & -b \\ 0 & 0 & 1 & 0 & 0 & b & -1 & -b \\ 0 & 0 & 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & b & 0 & -1-b \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

4.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= (1 - b) u_6 + u_7 + (-1 + b) u_8, \\ u_2 &= (1 - b) u_6 + b u_8, \\ u_3 &= -b u_6 + u_7 + b u_8, \\ u_4 &= u_6 + u_7 - u_8, \\ u_5 &= -b u_6 + (1 + b) u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + (1 - b) u_6 + u_7 + (-1 + b) u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d) .

4.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_7 :

$$u_7 = 1 + (-1 + b) u_6 + (1 - b) u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = x_5 = x_6 = (-1 + b) u_6 + (1 - b) u_8, \\ x_7 &= x_8 = (-2 + 2b) u_6 + (2 - 2b) u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 + b)u_6 - bu_8 &\not\equiv 0 \pmod{m}, \\ 1 - 2u_6 + u_8 &\not\equiv 0 \pmod{m}, \\ 1 + 2bu_6 + (-1 - 2b)u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_7 \neq t_8 &\iff 1 + (-1 + b)u_6 - bu_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 - 2u_6 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + 2bu_6 + (-1 - 2b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

5 Case IV: Subcase 2-1-2

5.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_3, c_6\}, \{c_2, c_7\}, \{c_4, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_4\}_1, \{c_5, c_7\}_1, \{c_6, c_8\}_1, \{c_1, c_5\}_2, \{c_2, c_7\}_2, \{c_3, c_6\}_2.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 4, 2, 7, 5, 1, 3, 6, 8(12121212):

$$\begin{aligned} &(-1 - 2b)u_1 + (1 + 2b)u_2 + (1 + 2b)u_3 + (-1 - 2b)u_4 \\ &+ (1 + 2b)u_5 + (-1 - 2b)u_6 + (-1 - 2b)u_7 + (1 + 2b)u_8 = 0. \end{aligned}$$

(2) 1, 5, 7, 2(2120):

$$(2 + b)u_1 + (-2 - b)u_2 + (-1 - 2b)u_5 + (1 + 2b)u_7 = 0.$$

(3) 3, 1, 5, 7, 2, 4(121210):

$$\begin{aligned} &(1 + 2b)u_1 + (-1 - 2b)u_2 + (1 - b)u_3 + (-1 + b)u_4 \\ &+ (-1 - 2b)u_5 + (1 + 2b)u_7 = 0. \end{aligned}$$

(4) 5, 1, 3, 6(2120):

$$(-1 - 2b)u_1 + (1 + 2b)u_3 + (2 + b)u_5 + (-2 - b)u_6 = 0.$$

(5) 7, 5, 1, 3, 6, 8(121210):

$$\begin{aligned} &(-1 - 2b)u_1 + (1 + 2b)u_3 + (1 + 2b)u_5 + (-1 - 2b)u_6 \\ &+ (1 - b)u_7 + (-1 + b)u_8 = 0. \end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{212} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & -1-b & b & 1 \\ 0 & 1 & 0 & -1 & 0 & 0 & b & -b \\ 0 & 0 & 1 & -1 & 0 & -1-b & 0 & 1+b \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

5.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_4 + (1 + b)u_6 - bu_7 - u_8, \\ u_2 &= u_4 - bu_7 + bu_8, \\ u_3 &= u_4 + (1 + b)u_6 + (-1 - b)u_8, \\ u_5 &= u_6 + u_7 - u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + (1 + b)u_6 - bu_7 - u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d).

5.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_4 :

$$u_4 = 1 + (-1 - b)u_6 + bu_7 + u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (1 + 2b)u_7 + (-1 - 2b)u_8, \\ x_5 &= x_6 = -2 - b + (2 + b)u_6 + (2 + b)u_7 + (-2 - b)u_8, \\ x_7 &= x_8 = -2 - b + (1 + 2b)u_6 + (2 + b)u_7 + (-1 - 2b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_7 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_6 - u_7 + u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 - b)u_6 - u_7 + (1 + b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_6 + bu_7 - bu_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 - b)u_6 + bu_7 &\not\equiv 0 \pmod{m}, \\ u_6 - u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned}
\text{a1: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\
\text{a2: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\
c1 &\iff 1 - u_6 - u_7 + u_8 \not\equiv 0 \pmod{m}, \\
c2 &\iff 1 + (-1 - b)u_6 - u_7 + (1 + b)u_8 \not\equiv 0 \pmod{m}, \\
c3 &\iff 1 - u_6 + bu_7 - bu_8 \not\equiv 0 \pmod{m}, \\
c4 &\iff 1 + (-1 - b)u_6 + bu_7 \not\equiv 0 \pmod{m}.
\end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

6 Case V: Subcase 2-1-7

6.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned}
M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\
M_1 &= \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\
M_2 &= \{\{c_1, c_5\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_4, c_8\}\}.
\end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_4\}_1, \{c_5, c_7\}_1, \{c_6, c_8\}_1, \{c_1, c_5\}_2, \{c_2, c_6\}_2, \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 3, 1, 5, 7(1212):

$$(1 + 2b)u_1 + (-1 - 2b)u_3 + (-1 - 2b)u_5 + (1 + 2b)u_7 = 0.$$

(2) 4, 2, 6, 8(1212):

$$(1 + 2b)u_2 + (-1 - 2b)u_4 + (-1 - 2b)u_6 + (1 + 2b)u_8 = 0.$$

(3) 3, 1, 2, 4(1010):

$$(-1 + b)u_1 + (1 - b)u_2 + (1 - b)u_3 + (-1 + b)u_4 = 0.$$

(4) 5, 1, 2, 6(2020):

$$(-2 - b)u_1 + (2 + b)u_2 + (2 + b)u_5 + (-2 - b)u_6 = 0.$$

(5) 7, 5, 1, 2, 6, 8(120210):

$$\begin{aligned}
&(-2 - b)u_1 + (2 + b)u_2 + (1 + 2b)u_5 + (-1 - 2b)u_6 \\
&\quad + (1 - b)u_7 + (-1 + b)u_8 = 0.
\end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{217} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & -1 & -1 & 2 \\ 0 & 1 & 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

6.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_4 + u_6 + u_7 - 2u_8, \\ u_2 &= u_4 + u_6 - u_8, \\ u_3 &= u_4 + u_7 - u_8, \\ u_5 &= u_6 + u_7 - u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + u_6 + u_7 - 2u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d).

6.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_7 :

$$u_7 = 1 - u_4 - u_6 + 2u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (-1 + b)u_6 + (1 - b)u_8, \\ x_5 &= x_6 = (-2 - b)u_4 + (2 + b)u_8, \\ x_7 &= x_8 = (-2 - b)u_4 + (-1 + b)u_6 + 3u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod{m}, \\ u_4 - u_8 &\not\equiv 0 \pmod{m}, \\ u_4 - bu_6 + (-1 + b)u_8 &\not\equiv 0 \pmod{m}, \\ u_4 + bu_6 + (-1 - b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_4 - u_6 + u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_4 \neq t_8 &\iff u_4 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a3: } t_7 \neq t_8 &\iff 1 - u_4 - u_6 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff u_4 + bu_6 + (-1 - b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff u_4 - bu_6 + (-1 + b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

7 Case VI: Subcase 2-3-5

7.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_6\}, \{c_5, c_7\}, \{c_4, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_2, c_4\}, \{c_3, c_7\}, \{c_6, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_6\}_1, \{c_4, c_8\}_1, \{c_5, c_7\}_1, \{c_1, c_5\}_2, \{c_2, c_4\}_2, \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 3, 1, 5, 7(1212):

$$(1 + 2b)u_1 + (-1 - 2b)u_3 + (-1 - 2b)u_5 + (1 + 2b)u_7 = 0.$$

(2) 6, 2, 4, 8(1212):

$$(1 + 2b)u_2 + (-1 - 2b)u_4 + (-1 - 2b)u_6 + (1 + 2b)u_8 = 0.$$

(3) 3, 1, 2, 4(1020):

$$(-1 + b)u_1 + (2 + b)u_2 + (1 - b)u_3 + (-2 - b)u_4 = 0.$$

(4) 5, 1, 2, 6(2010):

$$(-2 - b)u_1 + (1 - b)u_2 + (2 + b)u_5 + (-1 + b)u_6 = 0.$$

(5) 7, 5, 1, 2, 4, 8(120210):

$$\begin{aligned} &(-2 - b)u_1 + (2 + b)u_2 + (-1 - 2b)u_4 + (1 + 2b)u_5 \\ &+ (1 - b)u_7 + (-1 + b)u_8 = 0. \end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{235} = \begin{bmatrix} 1 & 0 & 0 & b & 0 & -1 - b & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & b & 0 & 0 & -1 & -b \\ 0 & 0 & 0 & 0 & 1 & -1 - b & -1 & 1 + b \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

7.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= -bu_4 + (1+b)u_6 + u_7 - u_8, \\ u_2 &= u_4 + u_6 - u_8, \\ u_3 &= -bu_4 + u_7 + bu_8, \\ u_5 &= (1+b)u_6 + u_7 + (-1-b)u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 - bu_4 + (1+b)u_6 + u_7 - u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d).

7.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_4 :

$$u_4 = 1 + b - bu_6 + (-1-b)u_7 + (1+b)u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (-2-b)u_6 + (2+b)u_8, \\ x_5 &= x_6 = -2-b + (1+2b)u_6 + (2+b)u_7 + (-1-2b)u_8, \\ x_7 &= x_8 = -2-b + (-1+b)u_6 + (2+b)u_7 + (1-b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1-b)u_6 - u_7 + (1+b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - bu_6 - u_7 + bu_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-2-b)u_6 - u_7 + (2+b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-2-b)u_6 + bu_7 + u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1-b)u_6 + bu_7 &\not\equiv 0 \pmod{m}, \\ u_6 + (-1-b)u_7 + bu_8 &\not\equiv 0 \pmod{m}, \\ u_7 - u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_4 \neq t_8 &\iff 1 + (-1-b)u_6 - u_7 + (1+b)u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a3: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 + (-2-b)u_6 + bu_7 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + (-1-b)u_6 + bu_7 \not\equiv 0 \pmod{m}, \\ \text{c3} &\iff u_6 + (-1-b)u_7 + bu_8 \not\equiv 0 \pmod{m}, \\ \text{c4} &\iff 1 + (-2-b)u_6 - u_7 + (2+b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c5} &\iff 1 - bu_6 - u_7 + bu_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

8 Case VII: Subcase 2-4-3

8.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_7\}, \{c_4, c_5\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_4, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_7\}_1, \{c_4, c_5\}_1, \{c_6, c_8\}_1, \{c_1, c_5\}_2, \{c_2, c_6\}_2, \{c_3, c_7\}_2.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 4, 5, 1, 3, 7, 2, 6, 8(12121212):

$$\begin{aligned} &(-1 - 2b)u_1 + (1 + 2b)u_2 + (1 + 2b)u_3 + (-1 - 2b)u_4 \\ &+ (1 + 2b)u_5 + (-1 - 2b)u_6 + (-1 - 2b)u_7 + (1 + 2b)u_8 = 0. \end{aligned}$$

(2) 1, 3, 7, 2(1210):

$$(1 - b)u_1 + (-1 + b)u_2 + (1 + 2b)u_3 + (-1 - 2b)u_7 = 0.$$

(3) 3, 1, 5, 4(1210):

$$(1 + 2b)u_1 + (1 - b)u_3 + (-1 + b)u_4 + (-1 - 2b)u_5 = 0.$$

(4) 5, 1, 3, 7, 2, 6(212120):

$$\begin{aligned} &(-1 - 2b)u_1 + (1 + 2b)u_2 + (1 + 2b)u_3 + (2 + b)u_5 \\ &+ (-2 - b)u_6 + (-1 - 2b)u_7 = 0. \end{aligned}$$

(5) 7, 2, 6, 8(1210):

$$(1 + 2b)u_2 + (-1 - 2b)u_6 + (1 - b)u_7 + (-1 + b)u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{243} = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & -1 - b & 1 + b \\ 0 & 1 & 0 & 0 & 0 & -1 & -1 - b & 1 + b \\ 0 & 0 & 1 & 0 & -1 - b & 1 + b & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 - b & 1 + b & 0 & -1 \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

8.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_5, u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_5 + (1+b)u_7 + (-1-b)u_8, \\ u_2 &= u_6 + (1+b)u_7 + (-1-b)u_8, \\ u_3 &= (1+b)u_5 + (-1-b)u_6 + u_7, \\ u_4 &= (1+b)u_5 + (-1-b)u_6 + u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_5 + (1+b)u_7 + (-1-b)u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d).

8.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_5 :

$$u_5 = 1 + (-1-b)u_7 + (1+b)u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = 1 + 2b + (-2-b)u_6 - 3bu_7 + (1+2b)u_8, \\ x_5 &= x_6 = (-1-2b)u_7 + (1+2b)u_8, \\ x_7 &= x_8 = (-1+b)u_6 + (-1-2b)u_7 + (2+b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} 1 + bu_6 + (-2-b)u_7 + u_8 &\not\equiv 0 \pmod{m}, \\ u_7 - u_8 &\not\equiv 0 \pmod{m}, \\ u_6 + bu_7 + (-1-b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 + bu_6 + (-1-b)u_7 &\not\equiv 0 \pmod{m}, \\ 1 - u_6 + (-1-b)u_7 + (1+b)u_8 &\not\equiv 0 \pmod{m}, \\ u_6 - u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_5 \neq t_6 &\iff 1 - u_6 + (-1-b)u_7 + (1+b)u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a3: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 + bu_6 + (-2-b)u_7 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff u_6 + bu_7 + (-1-b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c3} &\iff 1 + bu_6 + (-1-b)u_7 \not\equiv 0 \pmod{m}, \\ \text{c4} &\iff 1 - u_6 + (-1-b)u_7 + (1+b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

9 Conclusion

For Cases II–VII, the fundamental-cycle systems yield the displayed RREFs; with $u_1 = 1$, their pivot relations give conditions (b) and (d) , while the exact level-0 partition and pairwise-distinctness tests give conditions (a) and (c) . Thus the six alternatives in Theorem 36 are fully verified.